

# Nathaniel Martes

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## EDUCATION

*B.S. in Computer Science, Wilkes University, GPA: 3.95/4.00, Honors*

*Expected Graduation – May 2027*

## EXPERIENCE

*System Administrator Intern, Wilkes University*

*Jun 2024 – Current*

- Automated Ubuntu server administration tasks using Ansible, saving 1 hour during software updates.
- Received mentorship from senior system administrators during on-boarding to understand server design and tools.
- Configured Python toolchains on workstations using Ansible, making it easier for students to debug programs.
- Migrated LDAP database to LDAPS using Bash, Ansible, and Linux, ensuring heightened security for user credentials.

*Software Engineer Intern, NVIDIA*

*May 2026 – Sep 2026*

- Contributed to the open-source distributed object storage system OpenStack Swift, that manages 1+ EB of storage and supports autonomous vehicle AI workflows that create 160M+ requests per day.
- Developed and tested Python middleware replicating AWS S3 checksums, enabling data workflows handling 41 PB to leverage 90% faster checksum algorithms while ensuring API compatibility through Pytest unit and functional tests.
- Leveraged AI agents such as OpenAI's Codex and Claude in agentic workflows to increase development speed.
- Utilized Prometheus, Grafana, and OpenSearch to analyze and identify client requests at risk from upcoming updates.
- Automated provisioning and teardown of distributed storage benchmarking environments using Ansible, Python, and Linux, reducing deployment time and improving testing efficiency.
- Reviewed 20+ open-source patches, providing code design feedback to senior and non-senior engineers.
- Collaborated with cross-functional teams to relay software release changes and new features for users and deployers.

*Software Engineer Intern, VetsEZ*

*Sep 2025 – May 2026*

- Built a microservice with a team of developers using Java, Spring Boot, OData and Postman, which created a REST API for legacy SQL schemas that supported crucial platform modernization efforts.
- Utilized Python and Postman to automate regression testing of REST API remote producer call (RPC) execution, which saved the team 10 hours of manual regression testing.
- Integrated test cases into a GitHub Actions CI/CD pipeline for a SwaggerUI OpenAPI documentation generator using Java, successfully increasing generator code coverage to 91%.
- Optimized service efficiency by building a dedicated metadata caching layer using Java, Spring Boot, Docker, AWS ECS, and PostgreSQL, reducing startup latency for dependent APIs by 3 minutes.
- Cooperated with several developers using Microsoft Teams and Github Issues, improving code quality and coherence.

*Software Engineer Intern, Software Engineering Institute at Carnegie Mellon University*

*May 2025 – Aug 2025*

- Used Python, Git, Docker, and a REST API to create a vulnerability detector for installed software packages across all internally managed systems, cutting potential vulnerability detection time from 30 days down to 1 day.
- Corrected 4,400+ SQLite entries with Python, achieving 100% accuracy in quarterly software license usage reports.
- Built a Python and REST API tool to detect duplicate software license usage, saving 2 hours of manual data analysis.
- Integrated 80+ test cases for internal Atlassian suite applications into DevOps CI/CD pipelines using Python, Pytest, and REST APIs, preventing deployment of misconfigured Docker containers.
- Collaborated in an Agile environment using Jira to track project progress and participate in daily Stand-Up meetings, enhancing team communication and improving overall project understanding.

## PROJECTS

*AWS Affiliated Hackathon 1st Place, Interactive AI Reflection Therapy Web Application*

*Oct 2025 – Oct 2025*

- Created a serverless full stack application using JavaScript, React, HTML, Tailwind CSS, Python, Git, and DynamoDB that creates AI interactive narrative chapters for therapy and user resonance which placed 1st among 200+ participants.
- Utilized AWS Cloud Development Kit (CDK) to create infrastructure-as-code (IoC) which deployed a REST API Gateway and Lambda functions that interacted with a NoSQL database and a Large Language Model (LLM).
- Collaborated with 2 developers to design, develop, and deploy the complete application within a 24-hour period.

*J.P. Morgan Chase Affiliated Hackathon 1st Place, AI Image Disaster Analyzer*

*Mar 2025 – Mar 2025*

- Worked in a team of 4 to develop a full stack AI image analysis application for determining room disaster readiness using Python, Javascript, TypeScript, React, HTML, and CSS which placed 1st among 470+ participants.
- Built a REST API using FastAPI and Google Gemini to perform server side prompting and AI parsing.
- Established Git and Docker workflows which lead to better team efficiency under limited development time.

## LEADERSHIP

*Vice President, Mathematics, Physics, and Computer Science Club at Wilkes University*

*Jan 2025 – Current*

- Managed hackathons for attendees to hackathons in University of Delaware and Leigh University, achieving a 1st place hackathon win and university recognition.
- Created general announcements using Discord for club information, meetings, and university events for over 60+ members.

*Martial Arts Instructor, Dual Dragon Karate*

*May 2022 – Current*

- Led hundreds of classes teaching self-defense techniques and traditional forms to 15+ students.
- Handled different student age groups from 4 years old to 50+, ensuring students are satisfied with the class and that they are learning real world skills.

*Treasurer, Society of Professional Hispanic Engineers (SHPE) Wilkes Chapter*

*Dec 2025 – May 2026*

- Gave proposal for national convention funding to 15+ student government members, resulting in a full funding amount of \$3800+.
- Kept track of funds from events and fundraisers using Google Sheets, ensuring organization at every step.

*Public Relations Officer, Society of Professional Hispanic Engineers (SHPE) Wilkes Chapter*

*Jan 2025 – Dec 2025*

- Managed LinkedIn social media page for Wilkes Chapter, displaying student success, community involvement, general updates to over 850+ users.

## SKILLS

*Programming:* Python, Java, TypeScript, JavaScript, Golang, SQL

*Tools:* Pytest, Spring Boot, FastAPI, React, HTML, CSS, Git, Docker, Linux, CI/CD, AWS, PostgreSQL, Ansible, Kubernetes